

POWER BI PROJECT REPORT

Healthcare Data Preparation, Modeling and Dashboard Analysis

Name: Harini R Register Number: E25DA021

Project Overview

This project uses Microsoft Power BI to organize healthcare records and present useful information through an interactive report. The work includes preparing patient and billing data in Power Query, arranging tables for analysis, setting up relationships, and designing dashboard visuals to explore billing, patient demographics, medical conditions, admission dates, and insurance providers.

1. Creating the Dim_Patient Table

The patient information was prepared in Power Query by retaining the relevant patient fields and removing repeated records. An index was added and renamed as Patient ID to provide an identifier for patient records. The table includes patient name, age, gender, blood type, and age group. This organized table supports demographic filtering and helps connect patient details to billing data.

Name	Age	Gender	Blood Type	Age Group	Patient ID
Bobby Jackson	30	Male	B-	19 - 35	
Leslie Terry	62	Male	A+	Above	
Danny Smith	76	Female	A-	Above	
Andrew Watts	28	Female	O+	19 - 35	
Adrienne Bell	43	Female	AB+	36 - 60	
Emily Johnson	36	Male	A+	36 - 60	
Edward Edwards	21	Female	AB-	19 - 35	
Christina Martinez	20	Female	A+	19 - 35	
Jasmine Aguilar	82	Male	AB+	Above	
Christopher Berg	58	Female	AB-	36 - 60	
Michelle Daniels	72	Male	O+	Above	
Aaron Martinez	38	Female	A-	36 - 60	
Connor Hansen	75	Female	A+	Above	
Robert Bauer	68	Female	AB+	Above	
Brooke Brady	44	Female	AB+	36 - 60	
MS. Natalie Gamble	46	Female	AB-	36 - 60	
Halley Perkins	63	Female	A+	Above	
Mrs. Jamie Campbell	38	Male	AB-	36 - 60	
Luke Burgess	34	Female	A-	19 - 35	
Daniel Schmidt	63	Male	B+	Above	
Christina Bell	67	Female	A-	Above	

2. Preparing the Fact_Billing Table

The billing query was reduced to the fields needed for financial and date-based analysis: Date of Admission, Billing Amount, Discharge Date, and Insurance Provider. Keeping these columns in a separate table makes the billing information easier to analyze and reduces unnecessary fields in the reporting layer.

Week 5 E25DA021

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New Source

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Data source settings

Manage Parameters

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Advanced Editor

Manage

Choose Columns

Remove Columns

Keep Rows

Remove Rows

Sort

Split Column

Group By

Replace Values

Data Type: Text

Use First Row as Headers

Replace Values

Merge Queries

Append Queries

Combine Files

Combine

Queries [4]

healthcare_dataset

Dim_Patient

Dim_Admission

Fact_Billing

Table.SelectColumns(Source,("Date of Admission", "Billing Amount", "Discharge Date", "Insurance Provider"))

12

Date of Admission

100%

Valid

0%

Error

0%

Empty

12

Billing Amount

100%

Valid

0%

Error

0%

Empty

12

Discharge Date

100%

Valid

0%

Error

0%

Empty

12

Insurance Provider

100%

Valid

0%

Error

0%

Empty

1

1/31/2024

18856.28131

02-02-2024

Blue Cross

2

8/20/2019

33643.32729

26-08-2019

Medicare

3

9/22/2022

27955.09608

07-10-2022

Aetna

4

11/18/2020

37909.78241

18-12-2020

Medicare

5

9/19/2022

14238.31781

09-10-2022

Aetna

6

12/20/2023

48145.11095

24-12-2023

UnitedHealthcare

7

11/3/2020

19580.87234

15-11-2020

Medicare

8

12/28/2021

45820.46272

07-01-2022

Cigna

9

7/1/2020

50119.22279

14-07-2020

Cigna

10

5/23/2021

19784.63106

22-06-2021

UnitedHealthcare

11

4/19/2020

12576.79561

22-04-2020

Medicare

12

8/13/2023

7999.58688

05-09-2023

Medicare

13

12/12/2019

43282.28336

28-12-2019

Cigna

14

5/22/2020

33207.70663

19-06-2020

UnitedHealthcare

15

10/8/2021

40701.59923

13-10-2021

UnitedHealthcare

16

1/1/2023

12263.35743

11-01-2023

Blue Cross

17

6/23/2020

24499.8479

14-07-2020

UnitedHealthcare

18

3/8/2020

17440.46544

02-04-2020

Cigna

19

3/4/2021

18843.02302

14-03-2021

Blue Cross

20

11/15/2022

23762.20358

22-11-2022

Cigna

21

6/28/2023

42.51458855

02-07-2023

Blue Cross

Query Settings

PROPERTIES

Name

Fact_Billing

All Properties

APPLIED STEPS

Source

Removed Other Columns

4 COLUMNS, 999+ ROWS

Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 08:53

30°C Sunny

Search

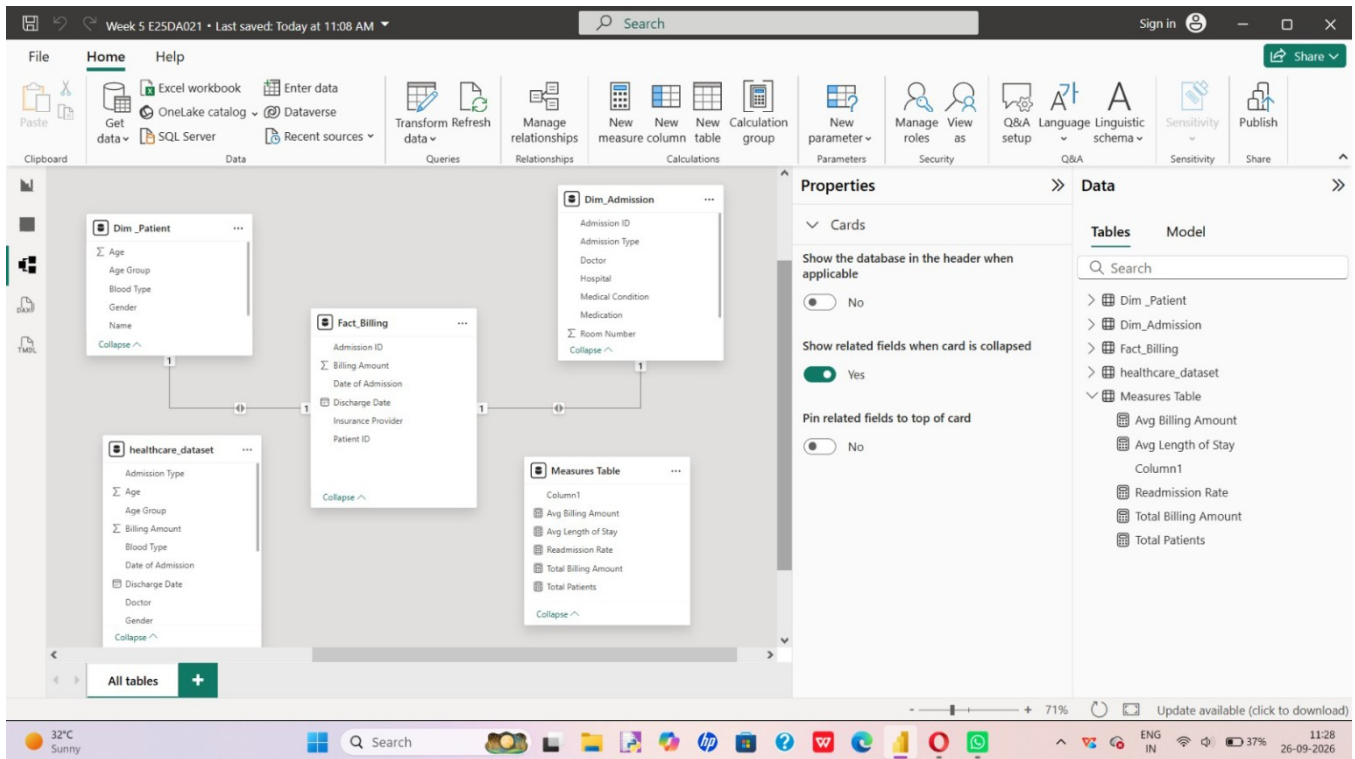
ENG IN

80%

08:53 26-09-2026

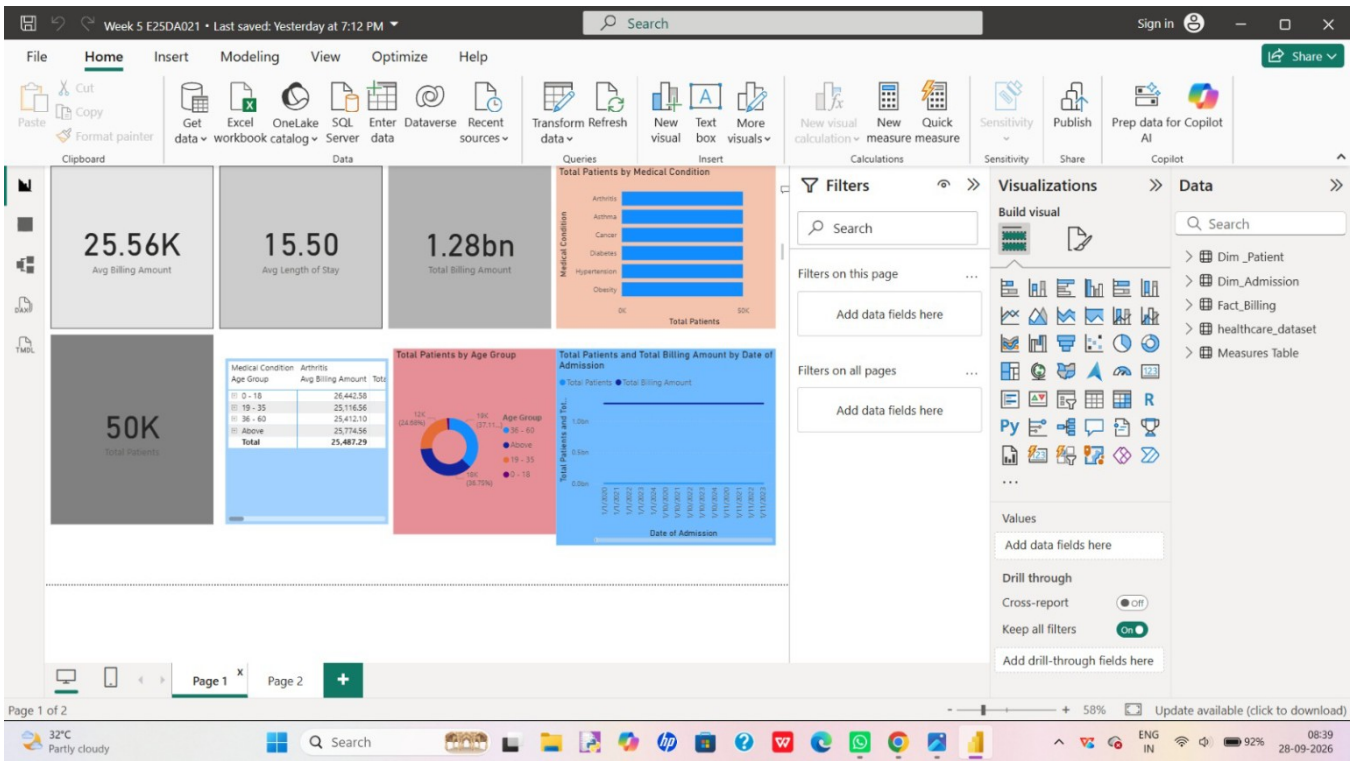
3. Establishing the Data Model

The model view displays the patient, admission, and billing tables, along with a separate measures table. Fact_Billing is linked to Dim_Patient through Patient ID and to Dim_Admission through Admission ID. These connections allow report visuals to use descriptive patient and admission information while summarizing billing records.



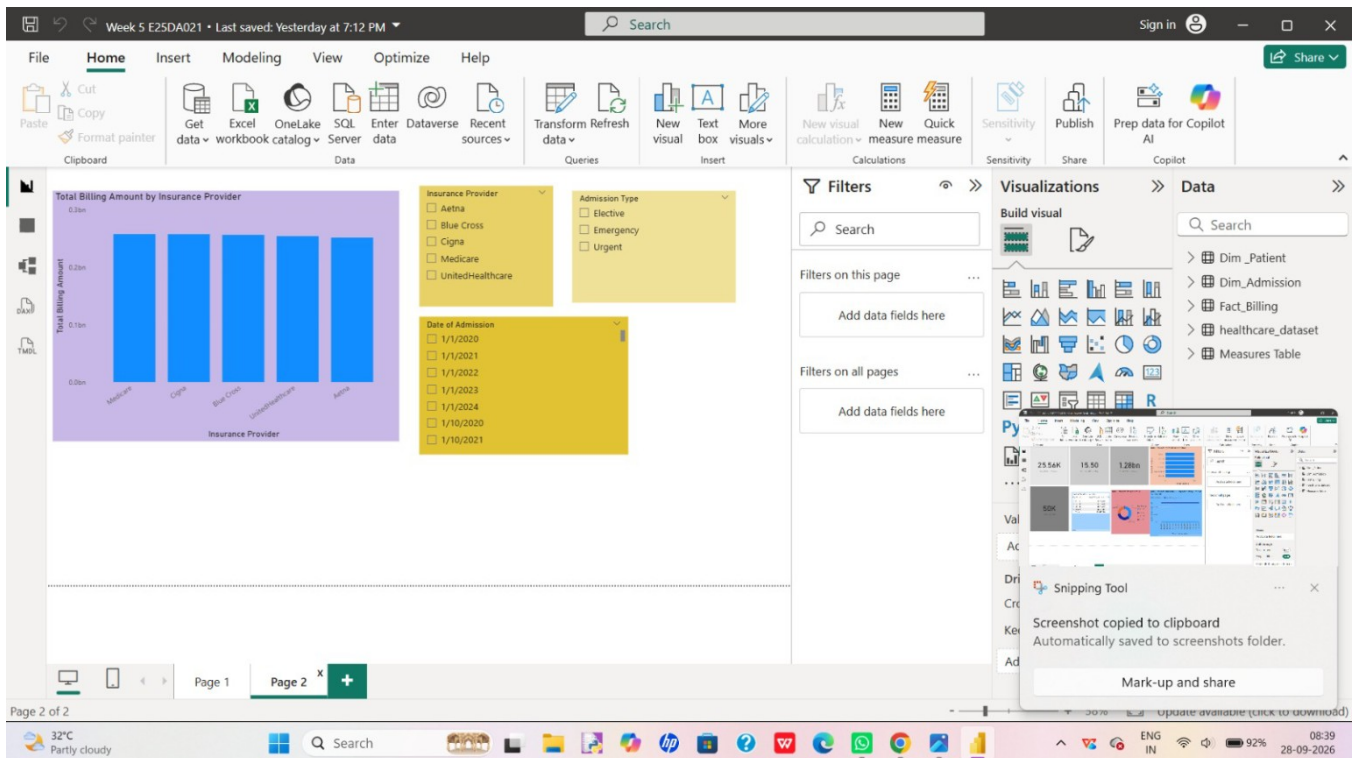
4. Dashboard – Patient and Billing Overview

The first dashboard page presents summary cards for average billing amount, average length of stay, total billing amount, and total patient records. Additional visuals compare billing by medical condition, patient counts by age group, and billing and patient totals over admission dates. A table provides a breakdown of billing amount across age groups and medical conditions.



5. Dashboard – Insurance and Admission Filters

The second dashboard page compares total billing amount across insurance providers using a column chart. Slicers for insurance provider, admission type, and admission date allow users to filter the report and inspect selected portions of the healthcare data.



Conclusion

The project demonstrates how Power Query, data modeling, and Power BI visuals can be combined to turn healthcare records into a structured analytical report. Separating patient, admission, and billing information and connecting them through identifier fields supports consistent filtering and analysis. The dashboard summarizes key billing and patient measures and provides slicers for exploring insurance, admission type, and admission dates.